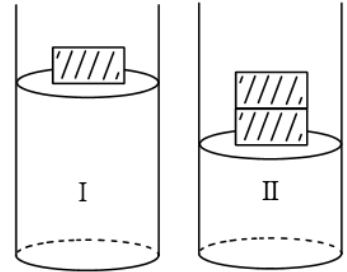


"By initialing below I agree to the following: I will not discuss this exam with others who may be taking it at a different time due to schedule conflicts, and that I will not copy off of another student's exam, both of which are considered cheating. I understand that if I am caught cheating on this exam I will receive a score of zero." Initials: Solution

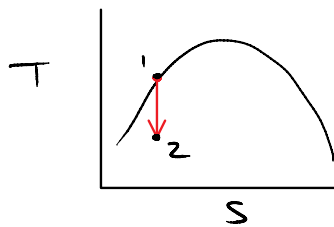
Justify your answer with a diagram, mathematically, or in words. Answers without justification will not receive credit.

- (3 Points) Both devices contain the same mass of liquid water and steam in vapor-liquid equilibrium. 10 kJ of heat is added to each.
 - The entropy remains constant in both devices.
 - The entropy for Device I goes up more.
 - The entropy for Device II goes up more.
 - The entropy goes up by the same amount in both devices.



$$\Delta S = \frac{Q}{T} \quad T_{II} > T_I \therefore \Delta S_{II} < \Delta S_I$$

- (3 Points) An adiabatic piston-cylinder device contains saturated liquid water. Sand is on top of the piston. Half of the sand is removed one grain at a time, halving the pressure. What is the final state of the system?
 - compressed liquid
 - $0 < x < 1$
 - $x = 1$
 - superheated vapor



The pressure drop is isentropic, putting the final state in the two-phase region.

- (3 Points) Two piston-cylinder devices contain ideal gas with 2 kg of sand resting on top. Initially, Device I contains 1 L of gas and Device II contains 2 L of gas. Sand is removed one grain at a time until 1 kg remains on each. The state change is isothermal.
 - Heat was not added to either device.
 - More heat was added to Device I.
 - More heat was added to Device II.
 - The same amount of heat was added to both devices.

Device II does more work because it contains twice the mass of gas. Therefore to keep them both isothermal, more heat must be added to Device II.

- (3 Points) Some liquid water is in equilibrium with its vapor in a piston-cylinder device. 100 kJ of heat is added, boiling off a little bit of water. The temperature does not change. Which is true?
 - $u_2 = u_1 + Q$
 - $h_2 = h_1 + Q$
 - $\Delta h = 0$ and $\Delta u = 0$
 - Both a. and b. are true

$$Q = \Delta u + P \Delta V = (u_2 - u_1) + (P_2 v_2 - P_1 v_1) = h_2 - h_1$$

$P_2 = P_1$

5. (6 Points) A volume of ideal gas at 300 kPa is expanded in the classroom to a pressure of 100 kPa and a volume that is three times as large.

I. The absolute temperature of the air ____.

- a. increases by a factor of three
- b. increases, but not by a factor of three
- c. decreases
- d. remains the same

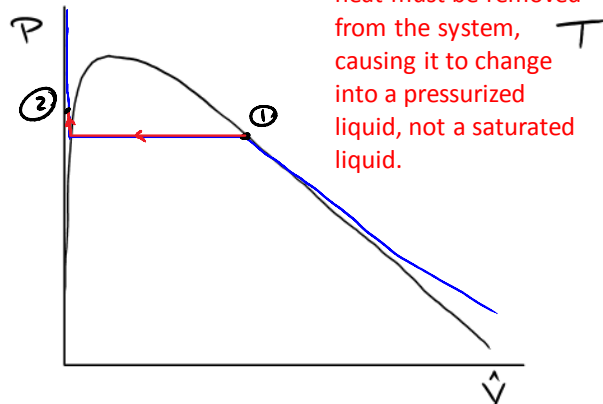
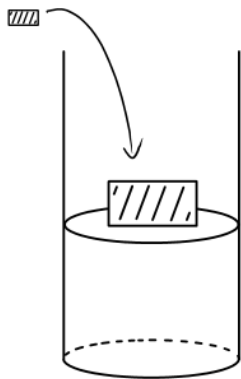
II.

- a. Heat flowed out of the air.
- b. Heat flowed into the air.
- c. Heat transfer was zero (adiabatic).

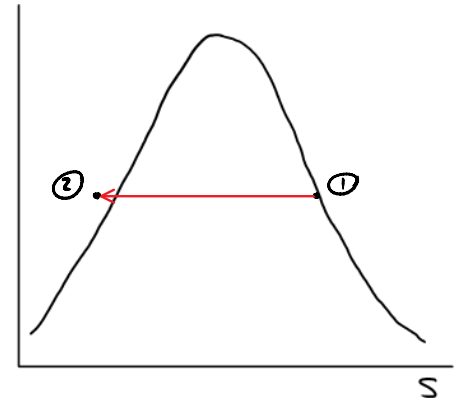
The air did boundary work.
To keep its internal energy (temperature) the same, heat had to flow into the air.

$$\frac{P_1 V_1}{P_2 V_2} = \frac{T_1}{T_2} \quad \frac{300}{100} \cdot \frac{1}{3} = \frac{T_1}{T_2} \rightarrow T_2 = T_1$$

6. (6 Points) Saturated water vapor ($x = 1$) is contained in a piston-cylinder device. A second small brick is added to the top of the device. The state transition is isothermal. Draw the state transitions on the P-V and T-s diagrams. What is the final state of the system?



To keep it isothermal, heat must be removed from the system, causing it to change into a pressurized liquid, not a saturated liquid.

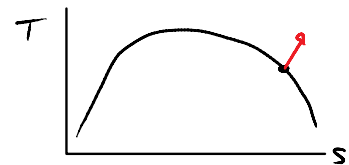


7. (3 Points) If the pressure of an ideal gas is reduced adiabatically and isentropically, its enthalpy [decreases / stays the same / increases].

Temperature decreases during isentropic, adiabatic expansion. If its temperature decreases, then the enthalpy of the gas decreases because enthalpy of an ideal gas is only a function of temperature.

8. (3 Points) If it is done adiabatically, it [is / is not] possible to form some liquid by squeezing a saturated vapor.

Squeezing a saturated vapor moves the state to the upper-right on a T-s diagram. There is no way to move into the two-phase region without removing heat.

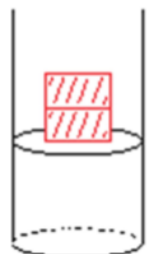


9. (3 Points) A volume of ideal gas is increased isothermally. The entropy of the gas [increases / remains the same / decreases].

$$\Delta S = C_v \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1} \quad V_2 > V_1 \therefore \Delta S > 0$$

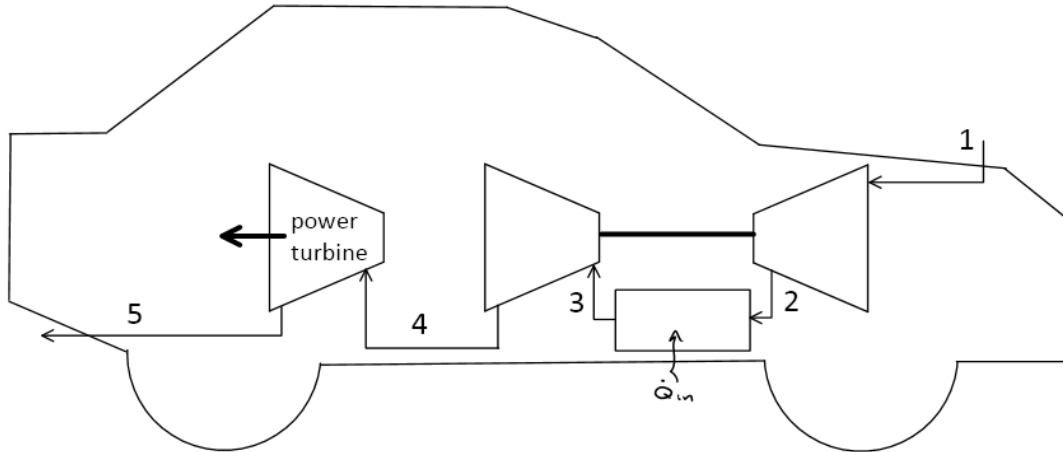
10. (3 Points) The top brick is removed and the ideal gas expands isothermally. The enthalpy of the gas [goes up / remains the same / goes down].

For an ideal gas, enthalpy is only a function of temperature. If the temperature remains the same, then the enthalpy also remains the same.



11. (30 Points) A car is designed to operate with a turbine engine. Air enters the car at a pressure and temperature of $P_1 = 100 \text{ kPa}$ and $T_1 = 20^\circ\text{C}$. The air is pressurized to $P_2 = 800 \text{ kPa}$. The temperature entering the power turbine is $T_4 = 2000^\circ\text{C}$. The air is exhausted at a pressure and temperature of $P_5 = 100 \text{ kPa}$ and $T_5 = 400^\circ\text{C}$. The engine generates 375 kW of power. The fuel has a heating value of 40 MJ/kg . Use $c_p = 1 \text{ kJ/kg}\cdot\text{K}$ and $k = 1.4$ for air.

- What is the engine's rate of fuel consumption?
- What is the engine's thermal efficiency?



$$\dot{m} h_4 = \dot{m} h_5 + \dot{w}_p$$

$$\dot{m} = \frac{\dot{w}_p}{c_p (T_4 - T_5)} = \frac{375 \text{ kW}}{1 \text{ kJ/kg}\cdot\text{K} (2000 - 400) \text{ K}} = 0.23 \text{ kg/s}$$

$$T_2 = T_1 \Gamma_p^{k-1/k} = (20 + 273) \text{ K}^{0.4/1.4} = 531 \text{ K} = 258^\circ\text{C}$$

$$\dot{w}_{in} = \dot{m} c_p (T_2 - T_1) = 0.23 \text{ kg/s} \cdot 1 \text{ kJ/kg}\cdot\text{K} \cdot (258^\circ\text{C} - 20^\circ\text{C}) = 55.75 \text{ kW}$$

$$\dot{w}_{out} = \dot{w}_{in}$$

$$\dot{w}_{out} = \dot{m} c_p (T_3 - T_4) \rightarrow T_3 = T_4 + \frac{\dot{w}_{out}}{\dot{m} c_p} = 2000^\circ\text{C} + \frac{55.75 \text{ kW}}{0.23 \text{ kg/s} \cdot 1 \text{ kJ/kg}\cdot\text{K}} = 2238^\circ\text{C}$$

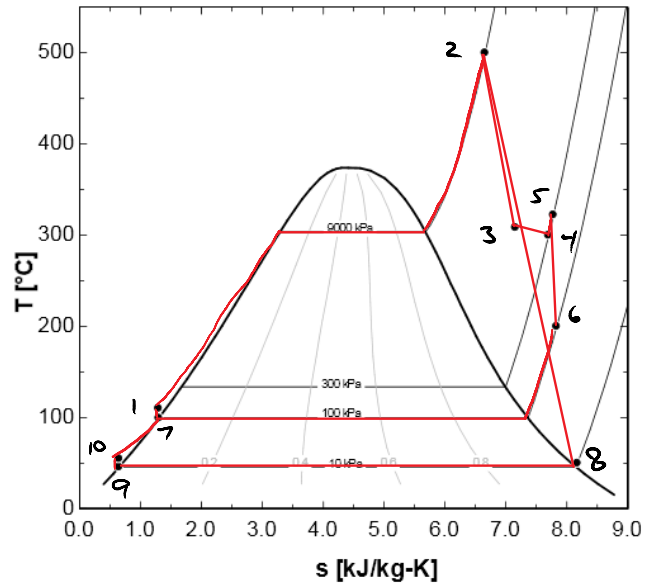
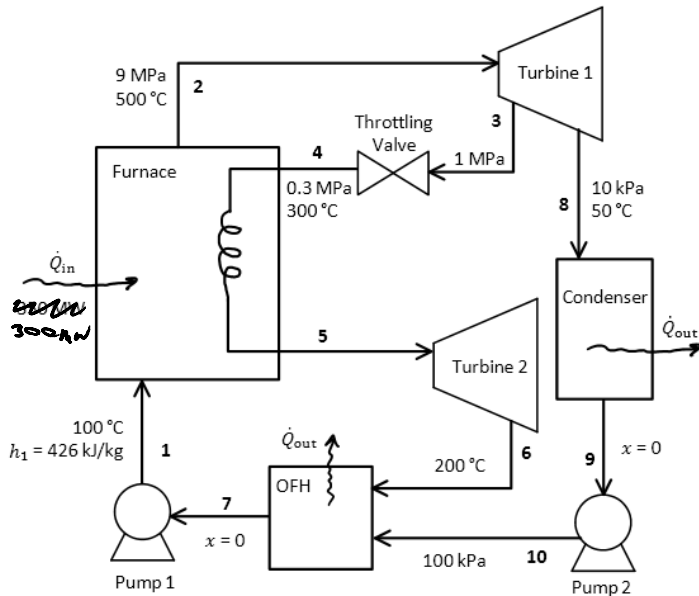
$$\dot{Q}_{in} = \dot{m} c_p (T_3 - T_2) = 0.23 \text{ kg/s} \cdot 1 \text{ kJ/kg}\cdot\text{K} (2238^\circ\text{C} - 258^\circ\text{C}) = 464 \text{ kW}$$

$$\dot{m}_{fuel} = \dot{Q}_{in} / HV = 464 \text{ kW} / 40000 \text{ kJ/kg} = 0.012 \text{ kg/s}$$

$$\eta_{th} = \dot{w}_p / \dot{Q}_{in} = 375 \text{ kW} / 464 \text{ kW} = 81\%$$

12. (30 Points) A flow diagram for a steam power plant is shown. Steam enters the furnace at a rate of 100 kg/s. 85% of the steam is bled from Turbine 1 and fed through the throttling valve. It is then reheated and fed to Turbine 2. The isentropic efficiencies of both turbines are less than 100%.

- Label and connect the dots on the T-s diagram, using arrows to indicate state transitions.
- What is the power output of Turbine 1?
- What is the power output of Turbine 2?



$$\dot{m}_2 h_2 = \dot{m}_3 h_3 + \dot{m}_8 h_8 + \dot{W}_{T1}$$

$$h_3 = h_4 = 3069.6 \text{ kJ/kg}$$

$$P_2 = 9 \text{ MPa} \left\{ \begin{array}{l} h_2 = 3387.4 \text{ kJ/kg} \\ T_2 = 500^\circ\text{C} \end{array} \right.$$

$$P_4 = 0.3 \text{ MPa} \left\{ \begin{array}{l} h_4 = 3069.6 \text{ kJ/kg} \\ T_4 = 300^\circ\text{C} \end{array} \right.$$

$$P_8 = 10 \text{ kPa} \left\{ \begin{array}{l} h_8 = 2592 \text{ kJ/kg} \\ T_8 = 50^\circ\text{C} \end{array} \right.$$

$$P_6 = 100 \text{ kPa} \left\{ \begin{array}{l} h_6 = 2875.5 \text{ kJ/kg} \\ T_6 = 200^\circ\text{C} \end{array} \right.$$

$$\dot{W}_{T1} = \dot{m}_2 h_2 - \dot{m}_3 h_3 - \dot{m}_8 h_8$$

$$= 100 \text{ kg/s} \cdot 3387.4 \text{ kJ/kg} - 85 \text{ kg/s} \cdot 3069.6 \text{ kJ/kg} - 15 \text{ kg/s} \cdot 2592 \text{ kJ/kg} = 38900 \text{ kW}$$

$$\dot{m}_1 h_1 + \dot{m}_4 h_4 + \dot{Q}_{in} = \dot{m}_2 h_2 + \dot{m}_5 h_5$$

$$h_5 = \frac{\dot{m}_1 h_1 + \dot{m}_4 h_4 + \dot{Q}_{in} - \dot{m}_2 h_2}{\dot{m}_5}$$

$$= \frac{100 \text{ kg/s} \cdot 426 \text{ kJ/kg} + 85 \text{ kg/s} \cdot 3069.6 \text{ kJ/kg} + 300000 \text{ kJ/s} - 100 \text{ kg/s} \cdot 3387.4 \text{ kJ/kg}}{85 \text{ kg/s}} = 3115 \text{ kJ/kg}$$

$$\dot{m}_5 h_5 = \dot{m}_6 h_6 + \dot{W}_{T2}$$

$$\dot{W}_{T2} = 85 \text{ kg/s} (3115 \text{ kJ/kg} - 2875.5 \text{ kJ/kg}) = 20400 \text{ kW}$$